UNIVERSITÉ GRENOBLE ALPES / ENSIMAG

Internship Report

Phage-Host Classification Using Conformal Prediction????

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Abstract

Acknowledgements

ENGLISH VERSION

Version française

Contents

1 Introduction

The rapid rise of antibiotic-resistant bacteria is one of the major challenges that global medicine is facing today. To overcome this issue, researchers are going to alternative solutions, one of which is phage therapy. Phage therapy relies on using the naturally occurring viruses to target and destroy the harmful bacteria.

However, To make this therapy a viable alternative, we first need to understand the basic biology of these viruses. Bacteriophages, or phages are viruses that infect bacterial hosts that and are one of the most abundant entities on the planet. Each phage contains a collection of structural and functional proteins that collectively determine the bacterial strains that the phage can infect - this property is called the host range of the phage.

These phages are highly host-specific: a phage which infects one bacterial host may have very little to no effect on another. This particular property of theirs has naturally generated interest in modelling and applications in the context of....... . Therefore a significant problem in the phage biology is the prediction of the bacterial hosts associated with a given phage. Accurate host-predictions can have applications in .....

complete sentence (treatment???)

find applications

The host specificity of phages is one of phage therapy's greatest advantage, but at the same time, it also brings a very significant practical challenge. While it allows us to target specific harmful bacteria without destroying the beneficial ones, it also causes probelms in finding exactly the appropriate phage for a specific infection. Doing this by only experimental screening can be time-consuming and expensive. This motivates the development of reliable computational models that are capable of predicting these phage-host interactions directly from the protein level information.

The existing computational tools, however have a very important limitation: they typically only produce point predictions, which are oftern represented as a scalar confidence score between 0 and 1 with no associated measure of uncertainty. This lack of having an uncertainty quantification is problematic in clinical contexts, where failing to identify a true host could mean missing a life-saving treatment. Consequently, to build trust in automated host prediction, we need a mathematical framework that is capable of quantifying its own predictive uncertainty and provides reliable confidence guarantees.

The primary objective of this report is to address this limitation by developing an uncertainty aware framework for phage-host prediction using **Conformal Prediction**. Conformal prediction provides a distribution-free framework that allows us to transform standard point predictions into prediction sets that are guaranteed to contain the true bacterial host with a user-defined confidence level.

In this project, we build our phage representations by aggregating collections of protein-level functional scores. This process naturally produced high-dimensional, structured score vectors, as different proteins and functions vary in their data availability. These score vectors also ofter contain missing or noisy information. To handle these challenges, we explored geometric conformal prediction methods that can construct reliable prediction sets even when dealing with incomplete biological data. As an intermediate step to validate our geometric approach in a controlled environment, we first evaluated the framework on a binary Gram-type classification task before moving on to the full multi-host prediction problem.

2 Background & State of the Art

In this section, we provide the biological, computational and mathematical background and context for the methods developed in this report. We will begin by introducing bacteriophages and the host-prediction problem in Section 2.1, which motivates the need for computational approaches to find a solution. We will then go over the already existing and state-of-the-art computational techniques in Section 2.2, progressing from the classical sequence similarity techniques to the much nore efficient and modern protein embeddings techniques. We will also look at the limitations that all the currently exisiting methods have in common, for example the absence of rigourous distribution-free uncertainity quantification. . In the remaining part of the section, 2.3 and 2.4 we will introduce the mathematical framework of conformal prediction and the score-based aggregation method of in [1], on which we will build our work.

define uncertainty quantification at somepoint in the report

2.1 Bacteriophages and Host-Prediction Problem

Bacteriophages, commonly referred to as phages, are viruses that specifically infect and replicate within bacteria. They are one of the most abundant biological entities on Earth with an estimated population of 10^{31} in the global biosphere [2], and therefore play a very important role in shaping the microbial communities and evolution. In the recent years the interest in bacteriophages has increased due to the growing threat of antimicrobial resistance and the development of phage therapy as an alternate treatment method to the traditional antibiotics [3].

One of the key characteristics of of phages is its host range, i.e. the set of bacterial hosts that the phage is capable of infecting. The host range of a phage is not a fixed property, it is determined by molecular interactions between the phage and the bacterial surface structures during the cycle of infection [2], [4]. Each phage attaches itself to the bacterial host, injects its viral genetic material which are able to replicate inside the host. Therefore, some of the phages only infect a few hosts (have a small host range) as they can only infect a limited number of bacterial strains while there are other which are capable of infecting a broad range of bacterias.

Identifying which phage infects which host, can have several applications both in environmental and clinical studies. In clinical, the host range helps us determine how suitable a phage is for designing the treatment for phage therapy (an alternate method to antibiotics). Similarly in environmental studies it is valuable to know the host being infected to understand the ecological interactions between viruses and microbial communities [4]. And it also improves our understanding of viral diversity and evolution.

Host range has its benefits, but at the same time it also brings its own obstacles. In phage therapy, we need to find precisely the phage that infects the given bacterial pathogen which is the cause of the infection. Therefore, we want to be able to identify the correct phage from the thousands of available ones and for that we need a fast reliable host prediction method. Traditionally, the method of determining the host range used to depend on laboratory experimentsm in which candidate phages were tested aginst a number of bacterial hosts; although this method was able to give accurate reliable results but it was extremely expensive and time consuming. Especially with the rapid increase in the phage genomes generated by modern technology, the number of possible phage-host interacions has increased astronomically, and this leads us to the need of having a compuational prediction method. We need a way which can complement the lab experiment by providing with a shorter list of candidate phages before we move on to expermiental validation [5] or even directly infer the host information directly from a given phage.

We can formulate this host prediction problem over a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $X \in \mathcal{X}$ be a high dimensional feature space representing the phage genomes and let $\mathcal{H} = \{h_1, \cdots, h_M\}$ be the finite set of possible bacterial hosts. In a single-host classification setting, the underlying data-generating process yields random pairs $(X, H) \sim \mathbb{P}_{X,H}$. In a general multi-host setting, the target is a non-empty subset of hosts $H^* \subseteq \mathcal{H}$, i.e., $H^* \in \mathcal{P}(\mathcal{H}) \setminus \{\emptyset\}$, where $\mathcal{P}(\mathcal{H})$ denotes the power set of $\mathcal{H}$. Our primary objective is to construct a set-valued prediction mapping:

$$C : \mathcal{X} \longrightarrow \mathcal{P}(\mathcal{H})$$

which maps a test phage $X_{n+1} \in \mathcal{X}$ to a prediction set $C(X_{n+1}) \subseteq \mathcal{H}$ that guarantees target coverage with a user-specified statistical confidence $1 - \alpha \in (0, 1)$.

2.2 Current Computational Approaches and Limitations

The problem of predicting which phage infects which host bacteria has been studied computationally for over a decade and the methods have significantly evolved during this time. Prediction methods have progressed from simple rule based alignemnt to deep representation learning. Here, we will summarize each family of methods and their underlying mathematical assumptions along with the common limitations that motivate us for this work.

2.2.1 Sequence Similarity Approaches

The earliest host prediction tools relied on sequence alignment between a phage and bacterial genomes, assuming that similarity between the two indicates infection compatibility. This was motivated by the known transfer of genetic material between the phages and their hosts through horizontal gene transfer.

One of the tools, HostPhinder [6], works by building a reference database of phage genomes with the known hosts. Then for a new phage in question, it assigns the host that is the host of its nearest neighbour under a k-mer similarity metric. Mathematically, let $v(g) \in \mathbb{R}^{4^k}$ be the normalised k-mer frequency vector of a genome g. So given a new test phage g_{test} and a reference database $\mathcal{D}_{\text{ref}} = \{(g_i, y_i)\}_{i=1}^{N}$ its predicted host is:

$$\hat{y} = \underset{y \in \mathcal{Y}}{\operatorname{argmax}} \left\{ \max_{g_i : y_i = y} \operatorname{sim}\left(\mathbf{v}(g_{\text{test}}), \mathbf{v}(g_i)\right) \right\}$$

where sim(.,.) is a similarity measure.

Another method, WIsH [7], uses a probabilistic approach by modelling the genome of each candidate host, $h \in \mathcal{H}$, as a variable order Markov chain with parameter vector, θ_h. It then computes the likelihood of the new phage genome sequence, $S = (a_1, \cdots, a_L)$ under each host model:

$$\ell(S \mid \theta_h) = \sum_{i=k+1}^{L} \log \mathbb{P}\left(a_i \mid a_{i-k}, \ldots, a_{i-1}; \theta_h\right)$$

The predicted host is $\hat{y} = \operatorname{argmax}_{h \in \mathcal{H}} \ell(S \mid \theta_h)$, which maximizes this likelihood.

Despite the fact that these methods are relitively simpler, they perform reasonably well when they have a phage reference that is closely related. But for a new phage that is not very common and is only a distant relative of the phages already there in the database, they degrade sharply. And this scenario is very common in metagenomic settings [5].

2.2.2 Protein Content Approaches

Instead of comparing and aligning the whole genomes, like the methods in 2.2.1 these methods break down the phage genome into its constituent proteins and then analyze their specific features from their functional or sequence properties. This approach is motivated by the fact that the proteins responsible for infecting a host are much more evolutionary conserved than the rest of the genome [8].

RaFAH [8] (Random Forest with protein Amino acid compositions for phage Host prediction) works by clustering all the known phage proteins into protein clusters using MMseqs2 [9]. It computes the amino acid composition vector of each cluster's representative sequence amd then trains a random forest classifier to predict the host genus from the protein cluster membership profile of the new phage. Let $x \in \{0,1\}^M$ be the binary indicator that encodes which of the M protein clusters are present in a given phage. Then the random forest maps this vector to the probability simplex over the host categories $\Delta^{|\mathcal{H}|-1}$:

$$f_{\text{RF}} : \{0,1\}^M \longrightarrow \Delta^{|\mathcal{H}|-1}, \quad \text{where } \hat{\mathbf{p}} = f_{\text{RF}}(\mathbf{x}), \quad \sum_{y \in \mathcal{H}} \hat{p}_y = 1$$

RaFAH is also relevant to this report, since we also work with similar protein based cluster representation and the contamination problem discussed in also arises exactly from this protein-cluster construction. add section reference

These techniques are more robust to sequence divergence than the sequence alignment methods 2.2.1, but they rely on the availibility of explicit functional annotations, which are missing for a large percentage of the already discovered phage genomes.

2.2.3 Protein Language Models and Embedding Based Approaches

The latest host prediction methods use protein language models (pLMs) [10, 11]. Similar to how large language models (LLMs) are trained on massive text datasets, pLMs are neural networks trained on millions of protein sequences through self-supervised learning.

A protein sequence of length n can be represented as $A = (a_1, \ldots, a_n) \in \mathcal{A}^n$, where $\mathcal{A}$ represents the standard amino acid alphabet of size 20. A is mapped by an encoder Φ_{pLM} (e.g., ProtT5-XL-UniRef50) [10] to a embedding matrix:

$$Z = \Phi_{\text{pLM}}(A) \in \mathbb{R}^{n \times d}$$

.

where d is the embedding dimension (typically 1024 or larger). This is followed by mean pooling along the sequence length which provides us with a fixed length embedding $z \in \mathbb{R}^d$.

$$\mathbf{z} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{Z}_{i,:}$$

This embedding vector, z, captures the sequence's biochemical, structural, and evolutionary properties without requiring an explicit sequence alignment. A primary foundational model in this domain is ProtTrans/ProtT5 [10], which is trained on hundreds of millions of UniRef protein sequences.

Several tools for phage protein protein annotation and host prediction have been developed using these embeddings. For instance, EMPATHI [12] uses a heirarchical classification pipeline to assigns functioal labels to phage proteins. It embeds the sequences using ProtT5 and performs nearest neighbour retreival witihin the embedding space, mapping them to a biological function ontology. SUBLYME [13] then extends this methodology for metagenomics which enables lysin discovery directly from environmemtal samples. These tools are directly involved in this work as rhe score matrix that is used as an input to this conformal framework is generated by a pipeline which is built on these annotations.

For host prediction, we can see in [14] that protein embeddings actually improve the predictions for the phage host interactions compared to the sequence-similarity baselines 2.2.1. In [15] Du et al. show that through some key protein embeddings LLMs can acheive high resolution host assignments. Such results support the theory that embedding represenations contain information that is sufficeint for an accurate host-prediction. But at the same time, all these methods output point predictions, a single score ot a single host label, and they do not provide us with any measure of uncertainity.

2.2.4 Limitations of Existing Methods and Need for Uncertainit Quantification

Despite the diversity and the empirical succcess of all the methods we have seen above, they all share a fundamental limitation, they all produce point predictions (a single predicted host or a single score) without quantifying the uncertainity of these predictions. For example, a score of $\hat{s}(s, y) = 0.87$ does not mean that there is a 87% chance of infection, it is just a relative ranking metric not a true probability of infection. More importantly none of these methods provide any mathematical guaranatee for coverage (that the true host would be included in the prediction set) with a given confidence level. This lack of uncertainity quantification can be a critical problem in some scenarios, like phage therapy, where picking the wrong phage for treatment can harm the patient.

Similarly, some phages are just genuinely difficult to classify which could be due to their host range having multiple hosts or simply the protein content may be insufficient to determine the exact host with confidence. In such cases a good method should be able to return all the plausible hosts rather than just forcing a singlem potentially wrong host. The existing methods do not have a formal way of doing this.

There are several existing frameworks for uncertainity quantification in machine learning, but most of them are not very well suited to this problem. For example, Bayesian approaches place a prior distribution over the model parameters and then compute a posterior given the observed data. Even though these approaches are supported theoretically, the issue with is that they require strong distributional assumptions that are difficult to justify fot biological score datasets. Also, computing or approximating the posterior is computationally expensive when we have a phage database with several hundred thousand genomes. Another problem is that the resulting uncertainity estimates depend heavily on the choice of the prior which introduces a degree of subjectivity which should be avoided in a clinical setup where the guarantee must be transparent and verifiable.

Other techniques, like the calibration methods include techniques like Platt scaling or temperature scaling.They adjust the raw output scores of the model so that they reflect the empirical probabilities better

$$\mathbb{P}\left(Y = y \;\middle|\; \hat{s}(X, y) = p\right) = p \quad \forall p \in [0, 1]$$

For example, they make sure that among all the predictions which are assigned the score of 0.8, around 80% should be correct. This is easy to apply on top of any existing models and is also compuationally cheap, but they do not provide us with any formal finite sample guarantees. The calibration is verified empirically on a validation set but it can degrade on new data with no bounds on how far the calibrated probabilities may deviate from the truth on the unseen test set.

These observations lead us to us the conformal prediction, [16], which offers exactly the properties required here. It is distribution-free, it requires no assumption on the underlying score distribution. It provides a finite-sample coverage guarantee rather than an asymptotic or empirical one. Its only requirement is exchangeability between calibration and test data. Also, it is computationally light, it requires only the computation of empirical quantiles rather than any iterative optimisation or sampling. Instead of replacing the currently existing methods, conformal prediction complements them by taking their raw scores inputs and then provides prediction sets with proven mathematical guanatees that can be proven and are detailes in . For these reasons, conformal prediction is the framework adopted in this report.

refer to the correct section

2.3 Conformal Prediction

Conformal prediction, initially developed by Vovk, Gammerman and Shafer [16] and then later made accessible by Angelopoulos and Bates [17], is a framework for constructing prediction sets that have a coverage guarantee. Unlike the methods seen in 2.2 it can provide us with a set of possible outcomes (not a point prediction), have a mathematical guarantee and yet requires no distributional assumptions on the data, except for the exchangibility condition. In this section we will develop the framework of conformal prediction and see

complete this sentence + define exchangeability

2.3.1 Mathematical Setup and Prediction Set

Let $(X, Y) \in \mathcal{X} \times \mathcal{Y}$ be a random input-label pair, where $\mathcal{X}$ is the feature space that represents our object of intererst (in our case it is the phage protein level score vectors) and $\mathcal{Y} = \{1, \cdots, K\}$ is the discrete label space (set of candidate hosts).

We assume that we have a calibration dataset, $D_{\text{cal}} = \{(X_1, Y_1), \cdots, (X_n, Y_n)\}$, of n exchangable data points drawn from an unknown joint distribution, $P_{X,Y}$. Then, for a new unlabelled test input, X_{n+1} our aim is to to construct a prediction set function $C : \mathcal{X} \to 2^{\mathcal{Y}}$ dependening on the data which maps the test input to a subset of candidate labels, i.e., $C(X_{n+1}) \subseteq \mathcal{Y}$, such that the true label falls in in the set with a defined coverage that is:

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

where $\alpha \in (0, 1)$ is the significance level. In our experiments we set $\alpha = 0.1$ so that the guarantee is 90%.

A prediction set function can provide us three qualitatively different outcomes for any new test point:

(i) $|C(x)| = 0$ (Abstension): this means that the model cannot find any plausibl host. This happens when the score profile of the text phage is very different from anything seen while calibrating.

(ii) $|C(x)| = 1$ (Confident Prediction): this means that exactly one host label is consistent with the calibration distribution at $1 - \alpha$ significance level.

(iii) $|C(x)| \geq 2$ (Unsure): this means that according to the model there are several plausible hosts.

The expected set size, $\mathbb{E}[|C(x)|]$ can help us determine the efficiency of the model. If the coverage is maintained, then the smaller the average set size the more efficient the model is, as it provides us with a more informative prediction. An ideal predictor should acheive a coverage $\geq 1 - \alpha$ with the smallest possible set size. This is the goal that we will work on in . refer to the correct section

2.3.2 Split conformal prediction

In this work, we have used split conformal prediction, in this the training data is split into two sets. The first set is the training set, it is used to fit a score model $\hat{f} : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We then define a nonconformity score function: $V : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ based on $\hat{f}$, here, $V(x, y)$ is used to calculate the error margins, it quantifies how unusual the pair (x, y) is relative to the training data. The calibration set (second set), is then used to evaluate the nonconformity scores:

$$V_i = V(X_i, Y_i), \quad \forall (X_i, Y_i) \in S_{cal}$$

Then by the exchangibility assumption, we have that the test score, $V_{n+1} = V(X_{n+1}, Y_{n+1})$ has rank which is uniformly distributed among $\{V_1, \cdots V_n, V_{n+1}\}$. Therefore, to guarantee that our error rate does not exceed α, we compute the empirical $(1 - \alpha)$ quantile of the calibration nonconformity scores, denoted as $\hat{q}$:

$$\hat{q} = \inf \left\{ q \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}(V_i \leq q) \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right\}.$$

Finally, for a new input, X_{n+1} the prediction set is constructed by collecting all the candidate labels in $y \in \mathcal{Y}$ whose nonconformity scores do not exceed this calibrated threshold:

$$C(X_{n+1}) = \{y \in \mathcal{Y} : V(X_{n+1}, y) \leq \hat{q}\}$$

2.4 The CSA Method

3 Methods

3.1 Problem Formalization

- Input: a phage represented by a score vector in $[0,1]^K$ where K is the number of functional dimensions
- Define the label space $\mathcal{Y}$ (finite set of classes)
- Goal: construct a prediction set $C(x) \subseteq Y$ with the conformal guarantee
- The one-envelope design: build one single envelope on all training phages under their true-label NC transformation, rather than separate envelopes per class

3.2 Non Convex envelope for classification

- In the regression setting of the original CSA paper, envelopes are typically convex because the score space is unbounded and the data does not cluster into separated groups
- In our classification setting, the score space is bounded in $[0,1]^K$ and — crucially — true NC vectors for different classes occupy distinct, well-separated regions of this space (example)
- In the classification, produce tighter prediction sets without sacrificing the coverage guarantee
- The guarantee still holds because it depends only on the exchangeability of the data and the quantile construction, not on the shape of the envelope

3.3 Score Transformation

- General formulation: for each class $y \in \mathcal{Y}$, define a transformation ϕ_y:$[0,1]^K \to [0,1]^K$ that maps the raw score vector to an conformal score vector such that a true infector of class y is near the origin
- Property we require: $phi_y(\mathbf{s})$ is small when $\mathbf{s}$ is consistent with label y, large otherwise
- we make a single unified envelope coherent across all classes

3.4 Envelope Constriction

3.4.1 CSA Two-Stage Calibration

- Training vectors are split into two disjoint halves S_1 and S_2 (50/50 random split).
- S_1 (shape discovery): used to learn the geometry of the envelope.
- S_2 (size scaling): used to calibrate the scaling factor $\hat{t}$ that determines the final size of the envelope.

- The two stages must use disjoint data to preserve the exchangeability argument underlying the coverage guarantee.

3.4.2 τ Computation

- Define $\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\}$, the minimum scaling factor needed to include point $\mathbf{s}$ in the scaled envelope E.

- The global scaling factor is then determined by the $(1-\alpha)$ quantile of the τ values over S_2:
$$\hat{t} = \tau_{(k)} \quad \text{where} \quad k = \lceil (|S_2| + 1)(1-\alpha) \rceil$$
where $\tau_{(k)}$ denotes the k-th smallest value of τ in S_2.

- Key contribution: for each of the three envelope geometries, $\tau(\mathbf{s})$ has a closed-form expression computable in a single pass — no iterative search required.

- Computational benefit: the entire calibration step reduces to one forward pass over S_2 followed by a sort.

- Contrast with the original CSA paper which uses iterative binary search.

3.4.3 Radial Method

3.4.4 Strip Method

3.4.5 1D

3.4.6 Class-Specific $\hat{t}$ Calculation

3.4.1 — CSA Two-Stage Calibration

3.5 $\hat{t}$ Calculation

- class conditional (why and how)

4 Data ....

4.1 Data Description

- The two files: protein-level score matrix (1,853,074 proteins × 40 functional score columns, values in [0,1], NaN when protein not annotated with that function) and metadata file (same 1,853,074 rows, columns: accession, host, host_type, split)
- Scale of the dataset: 18,474 unique phages after aggregation
- The split column: integer 0–4 encoding a genome-cluster-based partition — phages in the same genome cluster are always in the same fold, ensuring test phages are genuinely different from training phages at the sequence level
- Class distribution: gram-neg vs gram-pos vs unknown
- The NaN structure: NaN is structural missingness, not noise — a protein that was not annotated with a given function simply has no score for that function, which is different from a score of zero

4.2 Protein to Phage Aggregation

- The raw data is at the protein level but conformal prediction operates at the phage level — aggregation is needed
- NaN-aware mean pooling: for each phage and each functional score column, compute the mean across all proteins of that phage that have a non-NaN value for that column. Proteins with NaN for a given column are excluded from the mean for that column only
- Why not replace NaN with zero before pooling: treating an unannotated protein as having zero evidence would push the phage's aggregated score artificially toward zero, misrepresenting its biology
- After aggregation: result is a matrix of shape (number of phages) × 40, still with NaNs in cells where none of the phage's proteins were annotated with that function

4.2.1 NaN handling after aggregation

- After mean pooling, some phages still have NaN in certain columns -> this happens when none of their proteins were annotated with a particular function
- Strategy: fill remaining NaNs using column-wise medians computed strictly on the training set (never the test set, to prevent data leakage) - The median cap: any training median above 0.6 is capped at 0.5 -> this prevents an imputed value from falsely pushing a phage toward a particular class - any remaining NaNs after median imputation are filled with 0.5 (neutral value)

4.3 Train-Test Split

- Standard split: split 0 is the held-out test set, splits 1–4 form the training set
- The contamination problem: explain the protein cluster (pc) column -> a protein's cluster may include proteins from phages in different folds. The models that produced the scores were trained on protein clusters, so a protein from a training-fold phage may share a cluster with a test-fold protein, introducing subtle leakage
- The masking fix: the file all_random_pc_splits.pkl encodes for each protein and each (host, function) model which fold that protein was assigned to. Setting scores to NaN where the protein was not in the test set of the relevant model, before aggregation, removes the contaminated predictions. The existing NaN-aware mean pooling then handles these naturally -> this is the "missing value" framing

5 Experiments and Results

5.1 Synthetic Data Validation

5.1.1 Puspose and setup

- Why synthetic validation before real data: allows controlled verification that the coverage guarantee holds exactly, with known ground truth distributions
- General setup: simulate score vectors from known distributions, apply all three envelope methods, verify empirical coverage matches theoretical target $1 - \alpha$

5.1.2 2D Experimenats

- Three scenarios tested in 2D (K=2): - Uncorrelated scores: two independent uniform or Gaussian dimensions
- Correlated scores: scores with positive correlation -> tests whether the envelope adapts to the correlation structure
- Asymmetric predictors: one informative dimension and one uninformative dimension -> tests whether the envelope correctly ignores the noisy dimension
- For each scenario: show the envelope shape visually, report empirical coverage across multiple runs, confirm it matches $1 - \alpha$
- Key observation from 2D: the radial and strip envelopes are visibly non-convex and wrap tightly around the data, while collapsed 1D produces a simple interval

5.1.3 Extension to K Dimensions

- Repeat the experiments for $K \in 2, 5, 10, 20$ with a single host
- Show that all three methods maintain coverage $\geq 1 - \alpha$ across all dimensionalities
- Discuss the effect of increasing K on envelope quality: radial method becomes sparser (fewer S1 points per sector), strip method has more dimension pairs to condition on, collapsed 1D is unaffected
- Show that the analytical τ computation gives identical results to what iterative search would give, confirming correctness

5.1.4 Multiple Hosts

- Extend to a simulated multi-class setting with multiple hosts ($|\mathcal{Y}| = 6, K = 5$)
- Each host's true score distribution is a distinct cluster in $[0, 1]^K$
- Verify that the single unified envelope correctly assigns the true host to the prediction set with probability $\geq 1 - \alpha$ - Show 3D visualisation of the envelope for K=3

5.2 Gram Type Classification

5.2.1 Experimental Setup

- Data: split-0 phages only (80% calibration, 20% test), unknown host types excluded
- Parameters: $\alpha = 0.1, M = 200$ directions (radial), M=20 bins (strip), 10 independent runs to estimate variance
- The NC transformation used: keep scores as-is for one class, flip with 1 - s for the other

5.2.2 Results

- table: overall coverage, per-class coverage, average set size, empty set rate, size-2 rate, singleton accuracy std across 10 runs for all three methods
- Bar charts: set size distribution, overall and per-class coverage

5.3 Host Classification

6 Conclusion and Future Work

6.1 Summary.....

6.2 Limitations

- gram-pos coverage -

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